SYSDESIGN REFERENCE

Observability Tradeoff Matrix

A decision-oriented comparison of alternatives and failure modes.

Turn system behavior into actionable signals that support detection, diagnosis, and learning.

VERSION 1.0 REVIEW

REVIEWED THROUGH 2026-07-18

STATUS HUMAN REVIEW REQUIRED

SYSDESIGN / OBSERVABILITY

SYSDESIGN

Observability Tradeoff Matrix

CORE IDEA

Turn system behavior into actionable signals that support detection, diagnosis, and learning.

Reference architecture decision chain

01 Metrics	02 Logs	03 Traces	04 Slos

Decisions to defend

+ Metrics
+ logs
+ traces
+ SLOs
+ sampling
+ cardinality
+ and ownership.

Failure modes to test

+ Alert fatigue
+ high-cardinality cost
+ missing correlation
+ and dashboard-only monitoring.

Interview flow

- [ ] Clarify users, critical paths, and non-goals
- [ ] Estimate only scale that changes a decision
- [ ] Define API, data model, and source of truth
- [ ] Draw synchronous and asynchronous paths
- [ ] Pressure-test timeout, retry, overload, and recovery
- [ ] Close with tradeoffs and one alternative

Deep-dive drill

Observability Tradeoff Matrix

Probe	What a strong answer makes explicit
How does the design behave during a regional dependency failure?	Requirement, decision, cost, failure behavior, and observable metric.
Which metric reveals the first bottleneck?	Requirement, decision, cost, failure behavior, and observable metric.
What consistency does the user actually observe?	Requirement, decision, cost, failure behavior, and observable metric.
How would the design change at ten times the traffic?	Requirement, decision, cost, failure behavior, and observable metric.

Data questions

- Partition key
- Versioning
- Uniqueness
- Retention
- Reconciliation

Operations questions

- SLO
- Backlog
- Capacity
- Alert
- Failover test

DESIGN DISCIPLINE

For every box: which requirement justifies it, what cost does it introduce, and what happens when it fails?

PUBLISHING STATUS
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